

Cryptography - 17_10

Recall the GGM construction.

$$G : \{0, 1\}^\lambda \rightarrow \{0, 1\}^{2l}$$

$$G(k) = (\underbrace{G_0(k)}_{n \text{ bits}}, \underbrace{G_1(k)}_{n \text{ bits}})$$

We build $\mathcal{F} = \{F_k : \{0, 1\}^{n(\lambda)} \rightarrow \{0, 1\}^\lambda\}$ with $k \in \{0, 1\}^\lambda$ such that

$$F_k(x) = G_{x_n}(G_{x_{n-1}}(\dots G_{x_1}(k)) \dots)$$

For the proof, we use induction on input length $n(\lambda) = \text{poly}(\lambda)$.

Let $F'_k : \{0, 1\}^{n-1} \rightarrow \{0, 1\}^\lambda$, the PRF for input length $n - 1$ (so the last computational step), such that

$$F_k(x, y) = G_x(F'_k(y))$$

where $x \in \{0, 1\}; y \in \{0, 1\}^{n-1}$.

🔗 Lemma

If $\{F'_k\}$ is a PRF, then $\{F_k\}$ is a PRF family.

This Lemma implies the proof of GGM. In fact, for $n = 1$ GGM is built as:

$$F_k(x) = G_x(k) \quad x \in \{0, 1\}$$

x	$F_k(x)$
0	$G_0(k)$
1	$G_1(k)$

$\approx_c$

x	$R(x)$
0	\$
1	\$

Because $(G_0(k), G_1(k)) \approx_c U_{2\lambda}$

For the inductive step, take F'_k to be GGM construction on inputs of size $n - 1$ and note that F_k is exactly GGM on inputs of size n .

① Proof (Lemma)

Assuming that F' is a PRF, then F is a PRF.

We consider some experiments:

We have a module $\boxed{F_k}$ and an attacker $\mathcal{A}$. $\mathcal{A}$ sends x, y to F_k and he answers with z .

F_k computes, with $k \in U_\lambda$, $F_k(x, y) = G_x(F'_k(y))$.

This is $HYB_0(\lambda)$.

We want to make an hybrid step, using the security of the PRF.

We'll have a schema similar to above, but we have a box $\boxed{H}$ taking

$R' \leftarrow \mathcal{R}(\lambda, n - 1, \lambda)$.

Now, we have $H(x, y) = G_x(R'(y))$.

This is $HYB_1(\lambda)$.

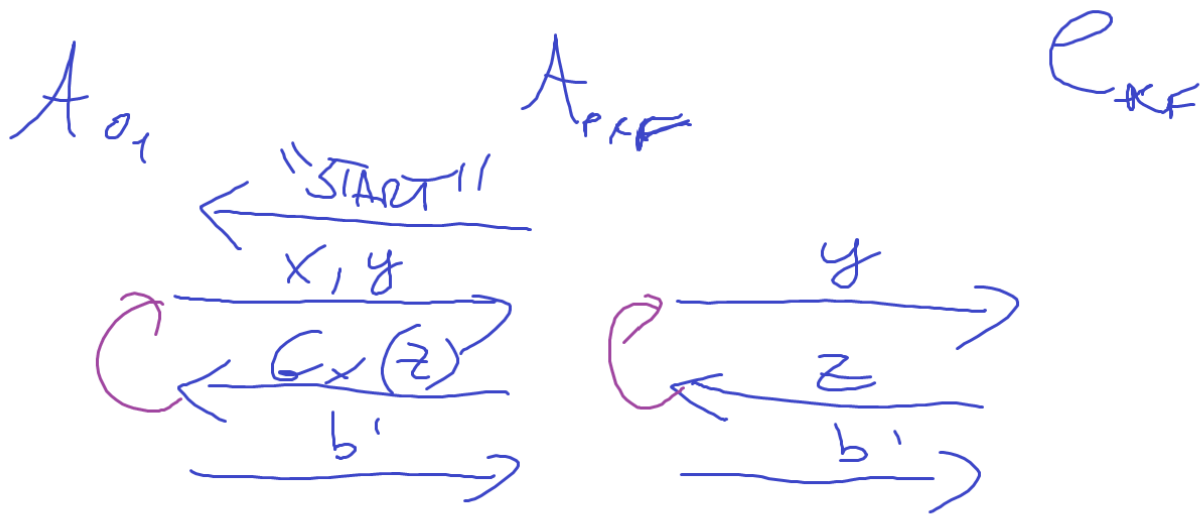
In the final game, we have a box $\boxed{R}$, with $z = R(x, y)$, and $R \leftarrow \mathcal{R}(\lambda, n, \lambda)$.

This is $HYB_2(\lambda)$.

Now, we first show $HYB_0(\lambda) \approx_c HYB_1(\lambda)$. This is true by reduction. Assume $\exists PPT \mathcal{A}_{01}$ telling apart HYB_0 and HYB_1 w.p. $\frac{1}{poly(\lambda)}$. Build reduction $\mathcal{A}_{prf}$ against $\{F'_k\}$.

So $\mathcal{A}_{01}$ is a machine making queries, so he sends x, y to $\mathcal{A}_{PRF}$ after $\mathcal{A}_{PRF}$ asks him to start. He replies with x, y (n bits). Then $\mathcal{A}_{prf}$ gives to $\mathcal{C}_{prf}$ just y , in fact it only accepts inputs of length $n - 1$. We get as a reply z . Now, we send to $\mathcal{A}_{01}$ $G_x(z)$. We can repeat this as many times as we want. And at the end we get b' that we

also give to the challenger.



Look: If $z = F'_k(y)$, then $G_x(z)$ is identical to what $\mathcal{A}_{01}$ receives in $\text{HYB}_0(\lambda)$.
On the other hand, if $z = R'(y)$ then $G_x(z)$ is identical to what $\mathcal{A}_{01}$ receives in $\text{HYB}_1(\lambda)$.

Next, we prove $\text{HYB}_1(\lambda) \approx_c \text{HYB}_2(\lambda)$.

We prove it by reduction, but first we use the following claim:

Claim

If $G : \{0, 1\}^\lambda \rightarrow \{0, 1\}^{2\lambda}$ is a PRG, then $\forall t(\lambda) = \text{poly}(\lambda)$:

$$(G(k_1), \dots, G(k_{t(\lambda)})) \approx_c (U_{2\lambda}, \dots, U_{2\lambda}) \equiv U_{2\lambda \cdot t(\lambda)}$$

where $k_1, \dots, k_t \in U_\lambda$

Let's assume the claim true and make the reduction: $\exists \text{PPT } \mathcal{A}_{12}$ telling apart $\text{HYB}_1(\lambda)$ and $\text{HYB}_2(\lambda)$ w.p. $\geq \frac{1}{\text{poly}(\lambda)}$. We build PPT $\mathcal{A}_{\text{prg}}$ breaking the above claim.

The challenger sends $z_1, \dots, z_q$. Then $\mathcal{A}_{\text{prg}}$ sends a start to $\mathcal{A}_{12}$, who sends back x, y .

$\mathcal{A}_{\text{prg}}$ answers by sending something, and do this for poly times, before having $\mathcal{A}_{12}$ sending b' .

Let $t(\lambda) = q(\lambda) = \# \text{queries made by } \mathcal{A}_{12}$.

This adversary $\mathcal{A}_{\text{prg}}$ receives q string (which could either be $G(k_1)$ (or so on), or they are uniform).

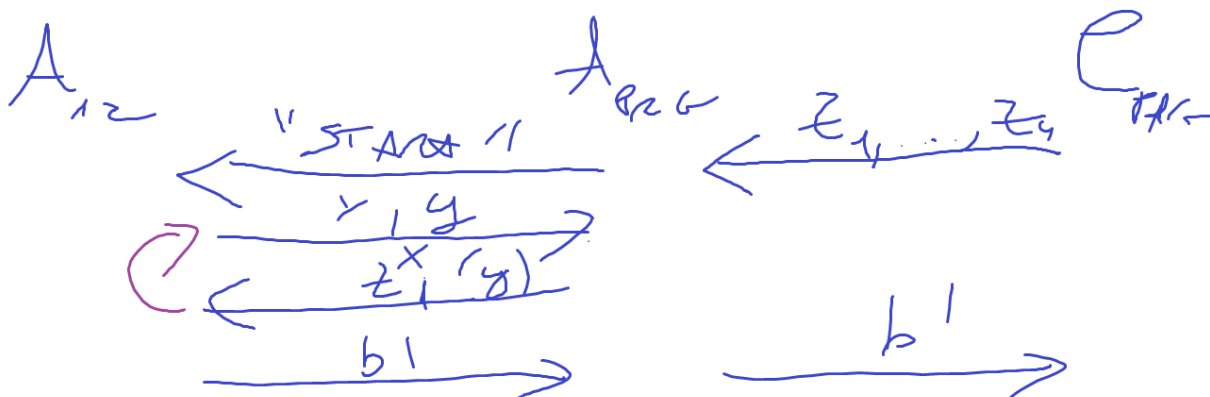
Now, after receiving x, y , what do we answer back?

Each of $z_i \in \{0, 1\}^{2\lambda}$ and we can think of it as

$$z_i = (\underbrace{z_i^0}_{\lambda \text{ bits}}, \underbrace{z_i^1}_{\lambda \text{ bits}}) \quad \text{two part of } \lambda \text{ bits}$$

Now, depending on the value of x , we either give z_i^0 or z_i^1 .

So, we put $z_i^x(y)$.



In the above reduction, $i(y)$ is the index of the sample z_i that was used when $\mathcal{A}_{12}$ asked already for x, y . If it never asked, use the next available z_i .

■

So now we know that $OWF \implies PRF$.

📌 Proof (Claim)

By reduction?

It only shows that $(U_{2\lambda}, \dots, U_{2\lambda}, G(k_t)) \approx_c (U_{2\lambda}, \dots, U_{2\lambda}, U_{2\lambda})$. This is basically 1st hybrid.

The second hybrid: $(U_{2\lambda}, \dots, U_{2\lambda}, G(k_{t-1}), G(k_t)) \approx_c (U_{2\lambda}, \dots, U_{2\lambda}, U_{2\lambda}, G(k_t))$.

Now, the challenger gives z to A_{prg} , which samples $k_t \in U_\lambda$, then it gives to A $U_{2\lambda}, \dots, U_{2\lambda}, z, G(k_t)$, and then we receive b' as an answer.

try to finish it up to just one reduction, here we have multiples